

22 Which statements correctly describe features of stem cells that are essential for their role in cell replacement and tissue repair?

1 After mitosis of stem cells, the daughter cells can either remain as stem cells or follow a developmental pathway that leads to the formation of specialised cells.

2 Stem cells are different to all other body cells because they retain all of the genetic information in their DNA throughout the life of the organism.

A small population of stem cells is retained in the body of adults throughout their life

3
time.

4 Stem cells have more telomeres than other body cells and this allows them to undergo an unlimited number of mitotic divisions.

A 1, 2, 3 and 4

B 1, 2 and 3 only

C 1 and 3 only

D 2, 3 and 4 only